INTERNATIONAL MONETARY FUND

FEDERATED STATES OF MICRONESIA

STAFF REPORT FOR THE 2021 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS

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Prepared by the staff of the International Monetary Fund and the International Development Association (IDA)[1,2]

Risk of external debt distress	High
Overall risk of debt distress	High
Granularity in the risk rating	Sustainable: Despite a continuous upward debt trajectory, the FSM's national assets including the FSM Trust Fund provide sufficient safeguards for debt sustainability.
Application of judgment	Yes: the forecast horizon informing mechanical risk signals is extended to 20 years to take account of the longer-term implications of a possible fiscal cliff in FY2023.

The Federated States of Micronesia (FSM) remains at high risk of debt distress under the Low-Income Country Debt Sustainability Framework (LIC DSF). The expected expiration of the U.S. Compact grants in FY2023 could result in a fiscal cliff. Under the baseline scenario without fiscal adjustment, the fiscal cliff would put debt on an upward trajectory from FY2024, with the external debt-to-GDP ratio reaching 44.2 percent in FY2031 and 81.1 percent in FY2041, and the public debt-to-GDP ratio reaching 45.1 percent in FY2031 and 82.3 percent in FY2041. As a result, the threshold of the present value of external debt-to-GDP ratio and the threshold of the public debt-to-GDP ratio are projected to be breached considering a 20-year horizon. While a mechanical

[1] The FSM's Composite Indicator of 1.67 indicates a weak debt-carrying capacity, based on the April 2021 IMF's World Economic Outlook (WEO) and the 2019 World Bank's Country Policy and Institutional Assessment (CPIA).

[2] This DSA has been prepared jointly by the IMF and World Bank, following the 2018 Guidance Note on the Bank-Fund Debt Sustainability Framework for Low Income Countries.

application of the DSF based on a 10-year forecast horizon would imply a moderate risk rating, the projected breach of the thresholds within a 20-year forecast warrants an assessment of high risk of external and overall debt distress. Debt is considered sustainable due to the availability of liquid assets, most importantly from the FSM trust fund. To lower the risk of debt distress and ensure debt sustainability, gradual fiscal consolidation, and steadfast structural reforms to promote private sector growth would be needed. The FSM's vulnerability to climate change and weather-related natural disasters constitutes a major risk and calls for strategies to strengthen climate change resilience.

PUBLIC SECTOR DEBT COVERAGE

1. This Debt Sustainability Analysis (DSA) covers debt owed by the national and state governments of the FSM (Text Table 1). As of May 2021, debt is primarily external and includes guaranteed debt from strategically important state-owned enterprises (SOEs). The legal framework in the FSM only allows for the central government on-lend to SOEs. There have been ongoing efforts by the authorities, supported by the FY2022 Performance and Policy Actions (PPAs) under IDA's Sustainable Development Finance Policy (SDFP), to enhance debt transparency, including reporting of both guarantees and non-guaranteed debt by SOEs. Against this background, the contingent liability stress test with default settings under the DSF is used to analyze public debt not covered by this DSA. The FSM uses the U.S. dollar as the legal tender and does not have a central bank. The external debt is defined based on the residency criteria.

Text Table 1. Coverage of Public Debt

	Subsectors of the public sector	Sub-sectors covered
1	Central government	X
2	State and local government	X
3	Other elements in the general government	
4	o/w: Social security fund	
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	
8	Non-guaranteed SOE debt	

	The country's coverage of public debt	The central, state, and local governments, government-guaranteed debt			
		Default		Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0	percent of GDP	0.0	
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2	percent of GDP	2.0	
4	PPP	35	percent of PPP stock	0.0	
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5	percent of GDP	5.0	
	Total (2+3+4+5) (in percent of GDP)			7.0	

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

BACKGROUND ON DEBT

2. The FSM's public sector debt declined to 15.4 percent of GDP at end-May 2021, from a peak of 30.7 percent of GDP in FY2009. Most of the debt is concessional and is contracted with official lenders (Text Table 2). As of May 2021, about 70.5 percent of the debt is from the Asian Development Bank (AsDB). The share of the U.S. Department of Agriculture declined to 18.1 percent, from 27 percent in the previous DSA, due to principal repayments and some write-offs. The remaining debt (11.4 percent of total) comprises the European Investment Bank (EIB), a commercial private lender, and domestic entities. All loans except from the EIB are denominated in U.S. dollars. Because the FSM has been running budget surpluses and

principal repayments have been proceeding, public sector debt shows a downward trajectory since FY2009. Total external debt service reached 6.7 percent of exports in FY2020 and is expected to remain broadly stable over the medium term. Nevertheless, it would rise in the long run, as most concessional loans enter repayment periods.

Text Table 2. FSM: Stock of Debt owed by the National and State Governments

	End-FY2018	May-21		
	In millions of U.S. dollars 1/	In millions of U.S. dollars	As a share of total debt	In percent of GDP
Total public debt	88.1	61.8	100.0	15.3
External	83.7	61.1	98.9	15.1
Multilateral	53.4	44.0	71.3	10.9
Asian Development Bank	52.5	43.6	70.5	10.8
European Investment Bank	0.9	0.5	0.8	0.1
Bilateral	20.7	11.2	18.1	2.8
U.S. Department of Agriculture	20.7	11.2	18.1	2.8
Commercial	9.6	5.8	9.4	1.4
Telecom vendor	5.0	3.2	5.2	0.8
Petrocorp	4.6	2.6	4.2	0.6
Domestic	4.4	0.7	1.1	0.2
Memorandum item:				
of which: SOE debt 2/	31.2	17.5	28.3	4.3

Source: FSM authorities and IMF staff estimates and calculations.
1/ Debt stock at end-FY2018 in the previous DSA amounted to US$ 76 million. The data revision of debt stock at end-FY2018 was mainly due to improved accounting of debt statistics.
2/ SOE debt comprises those of European Investment Bank, U.S. Department of Agriculture, Telecom Vendor, and Petrocorp.

BACKGROUND ON MACROECONOMIC FORECASTS

3. The fallout from the COVID-19 pandemic has resulted in an economic downturn in the near term, while fiscal and current account balances are expected to deteriorate from FY2024 onward as the Compact Agreement is expected to expire after FY2023 under the baseline projection (Box 1). Amid the COVID-19 pandemic, growth is expected to be negative during FY2020 and FY2021. A gradual recovery is expected from FY2022, driven by a pick-up in domestic activities due to the reopening of the economy. Potential growth is estimated to gradually decline from 0.6 percent in the medium term to 0.5 percent in the longer term, weighed down by the likely impact of rising climate change risks. Inflation is projected to converge to 2 percent over the longer term, in line with the U.S. inflation. Due to the possible expiration of grants under the Compact Agreement with the United States in 2023, the fiscal and current account balances are projected to turn sharply into deficits from FY2024. Despite the near-term disruptions, these projections are broadly in line with the previous DSA that accompanied the 2019 Article IV staff report for the FSM. The financing mix envisages a combination of external debt and domestic debt. The realism tools suggest that macroeconomic and fiscal assumptions are broadly reasonable (Figure 3 and 4)[3].

[3] The changes in the contribution of fiscal account balance to debt during the projection periods are due to significantly lower corporate tax revenue projections, as opposed to the large revenue windfalls seen in recent years, as well as the expected expiration of the Compact grants. In addition, a gradual recovery in FY2022 is expected to be driven by a broad recovery of domestic activities in line with a border reopening, which will be more than offset the negative growth impact of fiscal tightening in FY2022.

Box 1. Federated States of Micronesia: Key Macroeconomic Assumptions

- **Real GDP growth** is projected to fall to -1.8 percent and -3.2 percent in FY2020 and FY2021 respectively, amid the COVID-19 pandemic and related containment policies. The FSM has made good progress in vaccination efforts supported by the United States, with about 61 percent of the eligible population fully vaccinated as of September 13. Herd immunity will likely be achieved by end-2021. A slow recovery is expected for FY2022, driven by a gradual border reopening. Potential growth is estimated to be 0.6 percent in the medium term, in line with the FSM's growth performance over the last two decades. The estimate also accounts for the likely impact of rising climate change risks, as the FSM has experienced an increased frequency in natural disasters during this period. Based on Lee et al (2018)[1], the FSM's high vulnerability to natural disasters and rising climate change risks warrants further downward adjustment for GDP growth by 0.13 percentage point in the longer run.

- **Inflation** was moderate in FY2020 owing to low energy prices and downward price pressures from subdued economic activity. Due to higher import prices including for commodities, inflation is expected to rise from FY2022 on but would converge to 2 percent in the long term in line with inflation in the United States (the U.S. dollar is the FSM's legal tender). The GDP deflator is assumed to move in tandem with inflation.

- **The fiscal balance** is projected to be in a small surplus during FY2020-23. A sharp rise in COVID-19 related spending is expected to be fully financed by foreign grants and fishing license fees. Nonetheless, unless the Compact Agreement or parts of it are renewed, the Compact grants from the United States amounting to 22 percent of GDP will expire in FY2023 and be replaced by investment returns accruing to the Compact Trust Fund, projected at around 11 percent of GDP in FY2024. The baseline projection assumes no fiscal consolidation efforts between FY2021 and FY2024 leading to a possible fiscal cliff starting in 2024. The longer-term fiscal revenue is also subject to large fiscal revenue volatility related to investment returns and declining fishing license fees due to uncertain weather patterns.

- **The current account balance** is projected to remain in a small surplus prior to FY2023, amid the normalization of some one-offs related to corporate tax revenue windfalls during FY2017-2018. Nevertheless, the current account balance would turn into a deficit of around 4 percent of GDP on average during FY2024 and FY2041, resulting from the possible fiscal cliff.

- **FSM Trust Fund.** The FSM Trust Fund is accumulated with unspent revenues from fishing license fees, fiscal revenue windfalls, and occasional foreign grants. The FSM Trust Fund balance stood at US$307 million or about 75 percent of GDP at the end of FY2020. Under the baseline, the balance of the fund will moderately increase to around 130 percent of GDP by end-FY2041, as the rate of investment return is assumed to be higher than nominal GDP growth rate and a drawdown from the FSM Trust Fund is not considered. This assumption is in line with the authorities' intention to continue building the trust fund beyond FY2023 for the benefits of future generations.

- **Financing mix.** With the fiscal balance remaining in surplus and grants still abundant, no new debt incurrence is projected until FY2023. For instance, the World Bank has provided sizeable financial support to the FSM (as at end-June 2021, net commitments stood at around 55 percent of GDP and total disbursements were around 15 percent of GDP) and, consistent with the IDA grant allocation framework, such support has been provided as grants. Beyond this, the paths of external and public debt are determined by projected fiscal deficits, GDP growth, interest rates for new debt, and the financing mix for fiscal gross financing needs. The financing mix envisages external project loans financing 80 percent of gross financing needs, while the remaining 20 percent is assumed to be financed by domestic borrowing from banks. Such domestic borrowing would unlikely crowd out private sector credit, given the very low loan-to-deposit ratio in the FSM. New foreign grants are not assumed for the baseline, other than those already committed.

FSM: Macroeconomic Assumptions

	Current DSA (2021 Article IV)				Previous DSA (2019 Article IV)			
	2020	2021	2022-23	2024-41	2020	2021	2022-23	2024-39
Real GDP (y/y growth)	-1.8	-3.2	1.9	0.6	0.8	0.7	0.6	0.6
Inflation (y/y growth)	0.5	2.6	2.6	2.1	2.0	2.0	2.0	2.0
Fiscal balance (percent of GDP)	0.6	1.3	2.6	-5.0	6.7	6.4	4.9	-5.3
Current account balance (percent of GDI	3.0	1.2	0.4	-4.6	2.6	2.4	2.2	-4.2

Sources: FSM authorities and IMF staff estimates and calculations.

[1] See Lee and others, 2018, "The Economic Impacts of Natural Disasters in Pacific Island Countries: Adaptation and Preparedness," IMF Working Paper No. 18/108.

COUNTRY CLASSIFICATION AND DETERMINATION OF SCENARIO STRESS TEST

4. The FSM's debt carrying capacity is assessed as weak. The country's Composite Indicator (CI) index is 1.67, based on April 2021 WEO and 2019 CPIA (Text Table 3). The CI indicates a weak debt-carrying capacity, similar to the previous vintage classification. The DSA thresholds applicable for the FSM are: 30 percent for the present value (PV) of external debt-to-GDP ratio, 140 percent for the PV of external debt-to-exports ratio, 10 percent for the external debt service-to-exports ratio, 14 percent for the external debt service-to-revenue ratio, and 35 percent for the PV of public debt-to GDP ratio.

Text Table 3. FSM's Composite Indicator and Applicable Thresholds

Calculation of the CI Index

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.385	2.738	1.05	63%
Real growth rate (in percent)	2.719	0.573	0.02	1%
Import coverage of reserves (in percent)	4.052	0.759	0.03	2%
Import coverage of reserves^2 (in percent)	-3.990	0.006	0.00	0%
Remittances (in percent)	2.022	7.427	0.15	9%
World economic growth (in percent)	13.520	3.078	0.42	25%
CI Score			1.67	100%
CI rating			Weak	

Applicable Thresholds

Applicable	
External debt burden thresholds	
PV of debt in % of	
Exports	140
GDP	30
Debt service in % of	
Exports	10
Revenue	14

Applicable	
Total public debt benchmark	
PV of total public debt in percent of GDP	35

5. Given the severity and frequency of natural disasters in the FSM, a tailored stress test for a natural disaster shock is added to the standard set of stress test scenarios. In the natural disaster scenario, a one-off shock of 10 percentage points to debt-GDP ratio in FY2021 is assumed, with real GDP growth and exports growth lowered by 5.0 and 3.5 percentage points, respectively, in

the year of the shock.[4] Shocks related to contingent liabilities are considered, using the default shock parameters, while the scenario with exchange rate depreciation is not considered as the U.S. dollar is the FSM's legal tender.

DEBT SUSTAINABILITY

A. External Debt Sustainability Analysis

6. Under the baseline scenario, multiple debt indicators are projected to breach their respective thresholds after the standard projection horizon (2031). Due to the loss of grants and the resulting current account deficits post FY2023, the ratio of external debt to GDP is projected to rise from 13.6 percent in FY2023 to 81.1 percent in FY2041. As a result, the PV of external debt to GDP ratio breaches its threshold from FY2034 onward, while the ratio of the PV of external debt to exports breaches its threshold starting from FY2038. The external debt path is projected to deteriorate relative to the previous DSA, due to the lower current account surpluses resulting from the negative impact of the COVID-19 pandemic in the near term and a faster rise in external borrowing over the medium- and longer-term[5]. The debt trajectory under the historical scenario is significantly different from that of the baseline scenario, as it does not reflect the possible expiration of the Compact grants and widening fiscal deficits after FY2023.

7. Stress tests point to the vulnerability of the FSM's external debt dynamics to natural disasters and exports. The FSM is highly vulnerable to natural disaster shocks, which would raise the PV of external debt-to-GDP ratio above the threshold five years earlier than in the baseline and lead to an explosive debt path. Among the standardized tests, export shocks would result in the highest PV of external PPG debt-to-GDP ratio, breaching its threshold seven years after the shock.

B. Public Sector Debt Sustainability Analysis

8. Public sector debt follows closely the dynamics of external debt. Due to the fiscal cliff, the fiscal deficit is expected to remain sizable from FY2024 onward. The ratio of public debt to GDP is projected to rise from 13.6 percent of GDP in FY2023 to 82.3 percent in FY2041. Reflecting a gradually rising share of domestic debt in the government financing mix, public debt rises faster than external debt. The PV of the public debt-to-GDP ratio breaches the threshold under the baseline starting in FY2035. Stress tests indicate vulnerability of public debt dynamics to growth.

[4] The size of the shock on GDP growth (-5.0 percent), which is larger than the default value of the DSF framework (-1.5 percent), is based on cross-country regression results of Lee and others, 2018, "The Economic Impact of Natural Disasters in Pacific Island Countries: Adaptation and Preparedness," IMF Working Paper No.18/108.

[5] The previous DSA assumed a distribution from the FSM trust fund of about US$20 million per year after FY2030. Nevertheless, the current DSA does not make this assumption as the authorities do not intend any withdrawal.

RISK RATING AND VULNERABILITIES

9. The FSM remains at high risk of external and overall debt distress. The DSA's mechanical external and overall debt-distress ratings are moderate, as none of the DSA thresholds is breached within the standard 10-year horizon. Nevertheless, following the DSA guideline, the horizon for the FSM is extended to 20 years, accounting for sizeable fiscal deficits and sharp increases of debt post-FY2023. The projected breach of the thresholds within a 20-year forecast warrants an assessment of high risk of external and overall debt distress. The stress tests highlight the FSM's debt vulnerability to shocks in exports, growth, and natural disasters. The high-risk rating underscores the urgent need to implement fiscal adjustments and broader structural reforms to boost growth, develop a strategy for climate change resilience, and improve debt management capacity.

10. The debt dynamics is assessed to be sustainable. Despite a continuous upward debt trajectory, the FSM's national assets including the FSM Trust Fund provides an additional safeguard for debt repaying capacity and debt sustainability. Reflecting the authorities' intention, drawing from the FSM Trust Fund is not considered as part of the FSM's financing strategy. However, the trust fund remains a significant part of the national assets, thus facilitating the authorities' capacity to repay the public debt. Overall, steadfast fiscal adjustment and continued grant disbursements to support the FSM's large development needs would be necessary to keep debt on a sustainable path.

11. The future assessment of the risk of debt distress will critically depend on the outcome of the Compact renegotiation, including the size, composition, and duration of any grant support. The renewal of the Compact Agreements between the U.S. and the FSM will not be known before 2022. The continuation of grant support would help alleviate fiscal pressures and risks to long-term fiscal and external debt sustainability.

AUTHORITIES' VIEWS

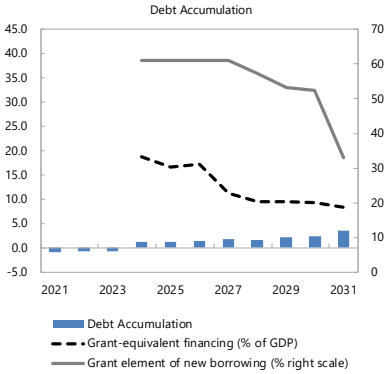
12. The authorities agreed with Staff's assessment that FSM remains at high risk of debt distress. They supported the use of a 20-year forecast horizon due to the pressure that the end of the Compact grants would put on the debt trajectory after FY2023. They emphasized that the government is unlikely to be able to borrow more than 10 percent of financing needs domestically after FY2023.

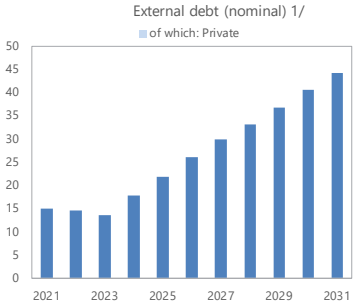
Table 1. Federated States of Micronesia: External Debt Sustainability Framework, Baseline Scenario, 2020-2041

(In percent of GDP, unless otherwise indicated)

	Actual	Projections								Average 8/	
	2020	2021	2022	2023	2024	2025	2026	2031	2041	Historical	Projections
External debt (nominal) 1/	**16.1**	**15.1**	**14.6**	**13.6**	**17.8**	**21.9**	**26.0**	**44.2**	**81.1**	**23.6**	**26.7**
of which: public and publicly guaranteed (PPG)	**16.1**	**15.1**	**14.6**	**13.6**	**17.8**	**21.9**	**26.0**	**44.2**	**81.1**	**23.6**	**26.7**
Change in external debt	-2.7	-1.0	-0.5	-1.0	4.2	4.1	4.1	3.7	3.1		
Identified net debt-creating flows	**-2.7**	**-0.6**	**-1.2**	**0.0**	**5.0**	**4.9**	**5.1**	**4.8**	**4.8**	**-3.3**	**3.3**
Non-interest current account deficit	**-3.4**	**-1.6**	**-1.6**	**-0.1**	**4.8**	**4.5**	**4.8**	**4.6**	**3.6**	**-3.3**	**3.0**
Deficit in balance of goods and services	42.4	47.3	43.2	43.0	40.1	39.4	39.2	37.5	34.7	**45.6**	**40.1**
Exports	24.0	24.2	31.7	32.2	32.1	32.1	32.1	32.1	32.1		
Imports	66.4	71.5	75.0	75.2	72.2	71.4	71.2	69.5	66.7		
Net current transfers (negative = inflow)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**0.0**	**0.0**
of which: official	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other current account flows (negative = net inflow)	-45.9	-48.9	-44.8	-43.1	-35.3	-34.8	-34.4	-32.9	-31.1	**-48.8**	**-37.1**
Net FDI (negative = inflow)	**0.1**	**0.1**	**0.1**	**0.1**	**0.1**	**0.1**	**0.1**	**0.1**	**0.1**	**0.1**	**0.1**
Endogenous debt dynamics 2/	**0.7**	**1.0**	**0.4**	**0.0**	**0.1**	**0.3**	**0.3**	**0.2**	**1.2**		
Contribution from nominal interest rate	0.5	0.5	0.5	0.4	0.4	0.4	0.4	0.4	1.5		
Contribution from real GDP growth	0.3	0.5	-0.1	-0.4	-0.2	-0.1	-0.1	-0.2	-0.4		
Contribution from price and exchange rate changes	-0.1	...	...	...	...	...	...	...	...		
Residual 3/	**0.0**	**-0.4**	**0.6**	**-0.9**	**-0.7**	**-0.8**	**-1.0**	**-1.1**	**-1.7**	**2.0**	**-0.8**
of which: exceptional financing	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators											
PV of PPG external debt-to-GDP ratio	**14.3**	**13.5**	**12.5**	**11.1**	**11.9**	**12.7**	**13.8**	**23.1**	**56.7**		
PV of PPG external debt-to-exports ratio	**59.4**	**55.8**	**39.2**	**34.3**	**36.9**	**39.7**	**43.1**	**71.9**	**176.7**		
PPG debt service-to-exports ratio	**6.7**	**6.7**	**5.6**	**5.3**	**5.1**	**5.0**	**4.3**	**3.4**	**8.1**		
PPG debt service-to-revenue ratio	**4.8**	**5.2**	**5.6**	**5.5**	**3.9**	**3.9**	**3.4**	**2.7**	**7.0**		
Gross external financing need (Million of U.S. dollars)	-7.2	0.2	0.9	7.5	29.8	29.7	30.6	32.2	44.8		
Key macroeconomic assumptions											
Real GDP growth (in percent)	-1.8	-3.2	0.6	3.2	1.9	0.8	0.6	0.6	0.5	**0.3**	**0.6**
GDP deflator in US dollar terms (change in percent)	0.5	2.6	2.6	2.6	2.6	2.5	2.4	2.0	2.0	**3.0**	**2.3**
Effective interest rate (percent) 4/	2.5	2.8	3.3	3.2	3.0	2.3	1.9	1.0	2.0	**2.6**	**1.9**
Growth of exports of G&S (US dollar terms, in percent)	-24.6	0.0	35.4	7.6	4.1	3.3	3.0	2.6	2.5	**4.8**	**6.1**
Growth of imports of G&S (US dollar terms, in percent)	-10.6	6.8	8.3	6.3	0.3	2.3	2.7	2.4	2.3	**1.3**	**3.4**
Grant element of new public sector borrowing (in percent)	...	53.7	61.0	61.0	61.0	61.0	61.0	32.9	28.7	**...**	**55.9**
Government revenues (excluding grants, in percent of GDP)	33.5	31.0	31.6	31.0	42.2	41.6	41.1	39.2	37.0	**35.4**	**38.0**
Aid flows (in Million of US dollars) 5/	146.2	169.5	155.7	153.4	97.4	90.8	95.6	46.9	56.1		
Grant-equivalent financing (in percent of GDP) 6/	...	41.9	36.8	34.2	18.7	16.7	17.2	8.4	8.1	**...**	**19.4**
Grant-equivalent financing (in percent of external financing) 6/	...	99.9	98.7	98.7	88.7	87.9	88.5	69.7	65.3	**...**	**86.9**
Nominal GDP (Million of US dollars)	407	404	417	442	462	478	492	561	720		
Nominal dollar GDP growth	-1.3	-0.8	3.2	6.0	4.6	3.3	3.0	2.6	2.5	**3.3**	**3.0**
Memorandum items:											
PV of external debt 7/	14.3	13.5	12.5	11.1	11.9	12.7	13.8	23.1	56.7		
In percent of exports	59.4	55.8	39.2	34.3	36.9	39.7	43.1	71.9	176.7		
Total external debt service-to-exports ratio	6.7	6.7	5.6	5.3	5.1	5.0	4.3	3.4	8.1		
PV of PPG external debt (in Million of US dollars)	58.1	54.6	52.0	48.9	54.8	60.8	68.0	129.5	407.9		
(PVt-PVt-1)/GDPt-1 (in percent)		-0.9	-0.7	-0.7	1.3	1.3	1.5	3.6	4.5		
Non-interest current account deficit that stabilizes debt ratio	-0.8	-0.7	-1.1	0.9	0.5	0.5	0.7	0.9	0.5		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	Yes





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + εα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, ε=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

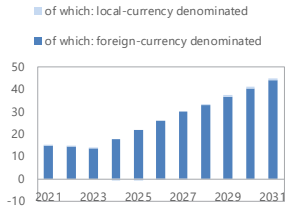
Table 2. Federated States of Micronesia: Public Sector Debt Sustainability Framework, Baseline Scenario, 2020-2041

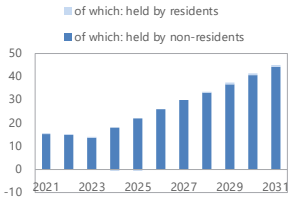
(In percent of GDP, unless otherwise indicated)

	Actual	Projections								Average 6/	
	2020	2021	2022	2023	2024	2025	2026	2031	2041	Historical	Projections
Public sector debt 1/	**16.5**	**15.3**	**14.6**	**13.6**	**17.1**	**21.3**	**25.6**	**45.1**	**82.3**	**23.8**	**26.8**
of which: external debt	**16.1**	**15.1**	**14.6**	**13.6**	**17.8**	**21.9**	**26.0**	**44.2**	**81.1**	**23.6**	**26.7**
Change in public sector debt	-3.1	-1.2	-0.7	-0.9	3.5	4.2	4.3	3.8	3.1		
Identified debt-creating flows	**-0.3**	**-0.9**	**-3.2**	**-3.1**	**3.5**	**4.3**	**4.4**	**3.8**	**3.1**	**-9.3**	**2.2**
Primary deficit	**-1.1**	**-1.8**	**-3.3**	**-2.8**	**3.7**	**4.5**	**4.6**	**4.4**	**3.4**	**-9.2**	**2.5**
Revenue and grants	69.3	72.8	67.7	64.6	57.2	54.7	54.8	45.8	43.4	**69.8**	**55.0**
of which: grants	35.9	41.9	36.1	33.5	15.0	13.1	13.7	6.6	6.4		
Primary (noninterest) expenditure	68.3	71.1	64.3	61.8	60.9	59.2	59.4	50.2	46.8	**60.7**	**57.5**
Automatic debt dynamics	**0.8**	**0.8**	**0.1**	**-0.3**	**-0.2**	**-0.1**	**-0.2**	**-0.6**	**-0.3**		
Contribution from interest rate/growth differential	0.5	0.8	0.1	-0.3	-0.2	-0.1	-0.2	-0.6	-0.3		
of which: contribution from average real interest rate	0.2	0.3	0.2	0.1	0.1	0.0	-0.1	-0.4	0.1		
of which: contribution from real GDP growth	0.4	0.6	-0.1	-0.5	-0.3	-0.1	-0.1	-0.2	-0.4		
Contribution from real exchange rate depreciation	0.2	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**-2.9**	**-0.2**	**2.5**	**2.2**	**0.0**	**-0.1**	**-0.1**	**0.0**	**0.0**	**8.0**	**0.4**
Sustainability indicators											
PV of public debt-to-GDP ratio 2/	**14.7**	**13.7**	**12.5**	**11.1**	**11.1**	**12.1**	**13.4**	**24.0**	**57.9**		
PV of public debt-to-revenue and grants ratio	**21.1**	**18.8**	**18.4**	**17.1**	**19.4**	**22.1**	**24.5**	**52.4**	**133.4**		
Debt service-to-revenue and grants ratio 3/	**3.1**	**2.6**	**2.9**	**2.7**	**2.9**	**1.9**	**1.8**	**3.5**	**7.6**		
Gross financing need 4/	1.1	0.1	-1.4	-1.1	5.3	5.5	5.5	6.0	6.7		
Key macroeconomic and fiscal assumptions											
Real GDP growth (in percent)	-1.8	-3.2	0.6	3.2	1.9	0.8	0.6	0.6	0.5	**0.3**	**0.6**
Average nominal interest rate on external debt (in percent)	2.5	2.8	3.3	3.2	3.0	2.3	1.9	1.0	2.0	**2.6**	**1.9**
Average real interest rate on domestic debt (in percent)	4.5	2.4	2.3	6.6	6.4	5.7	6.8	5.8	6.2	**0.0**	**5.6**
Real exchange rate depreciation (in percent, + indicates depreciation)	1.3	...	...	...	...	...	...	...	...	**-1.1**	**...**
Inflation rate (GDP deflator, in percent)	0.5	2.6	2.6	2.6	2.6	2.5	2.4	2.0	2.0	**3.0**	**2.3**
Growth of real primary spending (deflated by GDP deflator, in percent)	12.0	0.7	-8.9	-0.8	0.4	-2.1	0.9	0.2	-0.4	**0.9**	**-2.1**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	2.1	-0.6	-2.6	-1.8	0.2	0.2	0.3	0.6	0.3	**-12.6**	**-0.2**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**Yes**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central, state, and local governments, government-guaranteed debt . Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

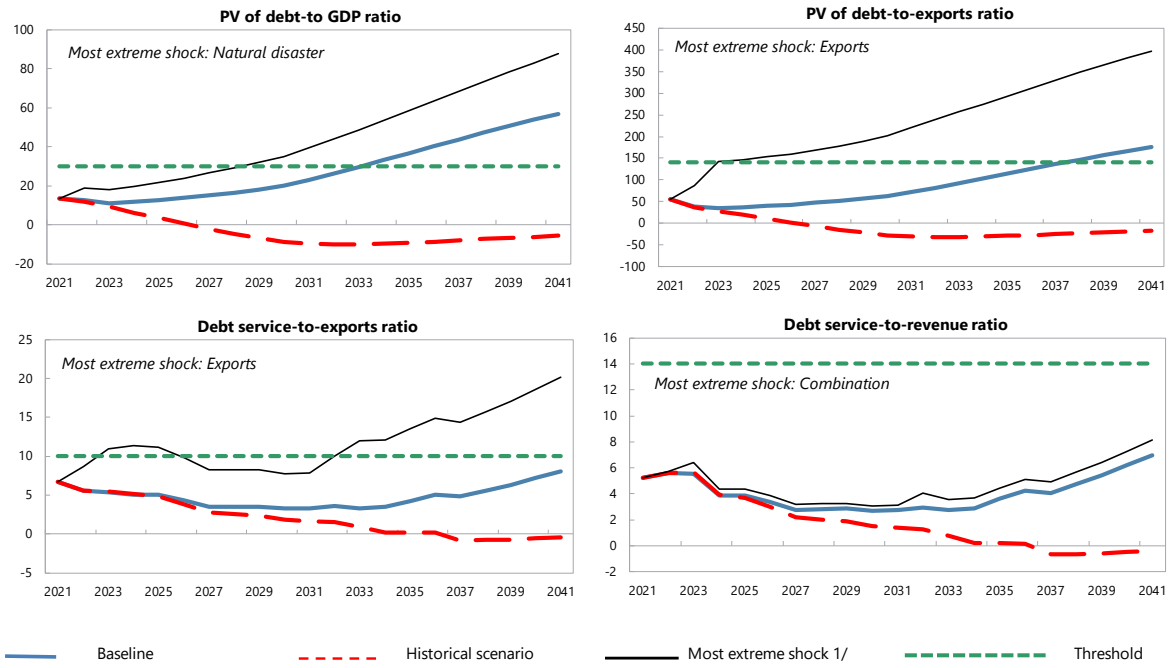
3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Federated States of Micronesia: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2021-2041



Customization of Default Settings	Size	Interactions
Standardized Tests	Yes	
Tailored Stress		
Combined CL	Yes	
Natural disaster	No	Yes
Commodity price	n.a.	n.a.
Market financing	n.a.	n.a.

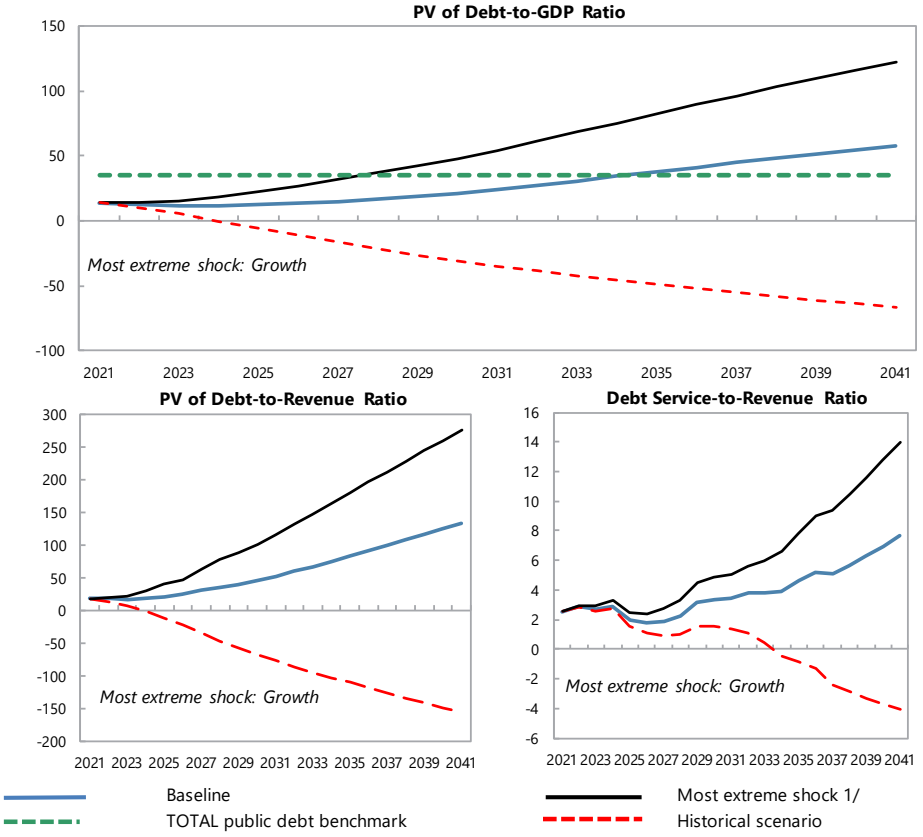
Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	1.0%	1.0%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	37	37
Avg. grace period	9	9

* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Figure 2. Federated States of Micronesia: Indicators of Public Debt under Alternative Scenarios, 2021-2041



Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	99%	99%
Domestic medium and long-term	0%	0%
Domestic short-term	1%	1%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	1.0%	1.0%
Avg. maturity (incl. grace period)	37	37
Avg. grace period	9	9
Domestic MLT debt		
Avg. real interest rate on new borrowing	7.9%	7.9%
Avg. maturity (incl. grace period)	3	3
Avg. grace period	2	2
Domestic short-term debt		
Avg. real interest rate	4.7%	4.7%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Federated States of Micronesia: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2021-2041

(In percent)

	Projections 1/											
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2041
PV of debt-to GDP ratio												
Baseline	14	12	11	12	13	14	15	16	18	20	23	**57**
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2021-2031 2/	14	12	9	6	3	0	-2	-5	-7	-9	-10	-6
B. Bound Tests												
B1. Real GDP growth	14	13	13	14	15	16	17	19	21	23	26	**65**
B2. Primary balance	14	16	19	19	20	22	23	24	26	28	**31**	**64**
B3. Exports	14	18	25	26	27	28	29	**31**	**33**	**35**	**39**	**69**
B4. Other flows 3/	14	13	11	12	13	14	15	17	18	20	23	**57**
B5. Depreciation	14	12	11	12	13	14	15	16	18	20	23	**57**
B6. Combination of B1-B5	14	17	15	16	17	18	20	21	23	25	28	**64**
C. Tailored Tests												
C1. Combined contingent liabilities	14	15	13	14	15	16	18	19	21	23	26	**60**
C2. Natural disaster	14	19	18	20	22	24	26	29	**32**	**35**	**39**	**87**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	30	30	30	30	30	30	30	30	30	30	30	30
PV of debt-to-exports ratio												
Baseline	56	39	34	37	40	43	47	51	57	62	72	**177**
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2021-2031 2/	56	37	28	19	10	1	-7	-14	-21	-28	-31	-18
B. Bound Tests												
B1. Real GDP growth	56	39	34	37	40	43	47	51	57	62	72	**177**
B2. Primary balance	56	51	58	61	64	67	72	76	82	88	98	**200**
B3. Exports	56	88	**142**	**147**	**153**	**160**	**169**	**178**	**189**	**202**	**221**	**397**
B4. Other flows 3/	56	39	35	37	40	44	48	52	57	63	72	**177**
B5. Depreciation	56	39	35	37	40	43	47	51	57	63	72	**177**
B6. Combination of B1-B5	56	75	42	71	76	81	88	94	103	113	128	**289**
C. Tailored Tests												
C1. Combined contingent liabilities	56	47	42	44	47	50	55	59	64	70	80	**186**
C2. Natural disaster	56	57	54	60	66	73	80	88	97	106	120	**266**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	140	140	140	140	140	140	140	140	140	140	140	140
Debt service-to-exports ratio												
Baseline	7	6	5	5	5	4	4	4	4	3	3	8
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2021-2031 2/	7	6	5	5	5	4	3	3	2	2	2	0
B. Bound Tests												
B1. Real GDP growth	7	6	5	5	5	4	4	4	4	3	3	8
B2. Primary balance	7	6	6	6	5	5	4	4	4	4	4	10
B3. Exports	7	9	**11**	**11**	**11**	10	8	8	8	8	8	**20**
B4. Other flows 3/	7	6	5	5	5	4	4	4	4	3	3	8
B5. Depreciation	7	6	5	5	5	4	4	4	4	3	3	8
B6. Combination of B1-B5	7	8	9	8	8	7	6	6	6	5	6	**14**
C. Tailored Tests												
C1. Combined contingent liabilities	7	6	5	5	5	4	4	4	4	3	3	8
C2. Natural disaster	7	6	6	6	6	5	4	4	4	4	4	9
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	10	10	10	10	10	10	10	10	10	10	10	10
Debt service-to-revenue ratio												
Baseline	5	6	6	4	4	3	3	3	3	3	3	7
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2021-2031 2/	5	6	6	4	4	3	2	2	2	1	1	0
B. Bound Tests												
B1. Real GDP growth	5	6	6	4	4	4	3	3	3	3	3	8
B2. Primary balance	5	6	6	4	4	4	3	3	3	3	3	8
B3. Exports	5	6	6	5	5	4	4	4	4	3	3	10
B4. Other flows 3/	5	6	6	4	4	3	3	3	3	3	3	7
B5. Depreciation	5	6	6	4	4	3	3	3	3	3	3	7
B6. Combination of B1-B5	5	6	6	4	4	4	3	3	3	3	3	8
C. Tailored Tests												
C1. Combined contingent liabilities	5	6	6	4	4	3	3	3	3	3	3	7
C2. Natural disaster	5	6	6	4	4	4	3	3	3	3	3	8
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	14	14	14	14	14	14	14	14	14	14	14	14

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Federated States of Micronesia: Sensitivity Analysis for Key Indicators of Public Debt, 2021-2041

(In percent)

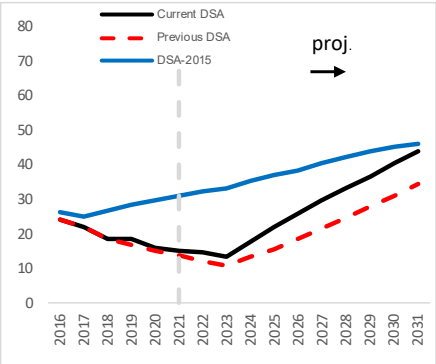
	Projections 1/											
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2041
PV of Debt-to-GDP Ratio												
Baseline	14	12	11	11	12	13	15	17	19	21	24	**58**
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2021-2031 2/	14	10	6	-1	-6	-11	-17	-22	-26	-31	-35	-67
B. Bound Tests												
B1. Real GDP growth	14	14	16	19	23	27	32	**37**	**42**	**48**	**54**	**122**
B2. Primary balance	14	16	19	19	20	21	23	25	27	29	32	**65**
B3. Exports	14	18	24	24	25	26	28	30	32	34	**38**	**67**
B4. Other flows 3/	14	13	11	11	12	14	15	17	19	21	24	**58**
B5. Depreciation	14	12	11	11	12	13	15	17	19	21	24	**58**
B6. Combination of B1-B5	14	17	16	14	15	17	18	20	22	25	28	**64**
C. Tailored Tests												
C1. Combined contingent liabilities	14	15	13	13	14	16	17	19	21	23	27	**61**
C2. Natural disaster	14	19	18	19	21	24	26	29	33	**36**	**40**	**89**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
TOTAL public debt benchmark	35	35	35	35	35	35	35	35	35	35	35	35
PV of Debt-to-Revenue Ratio												
Baseline	19	18	17	19	22	24	31	36	41	45	52	133
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2021-2031 2/	19	14	8	(1)	(11)	(21)	(34)	(46)	(57)	(68)	(76)	(155)
B. Bound Tests												
B1. Real GDP growth	19	20	23	31	40	48	64	78	90	102	116	276
B2. Primary balance	19	24	29	33	36	39	47	53	58	63	71	151
B3. Exports	19	26	37	41	45	48	57	64	69	74	82	155
B4. Other flows 3/	19	19	17	20	22	25	31	36	41	46	53	134
B5. Depreciation	19	18	17	19	22	24	31	36	41	45	52	133
B6. Combination of B1-B5	19	25	24	25	28	30	37	43	48	53	60	145
C. Tailored Tests												
C1. Combined contingent liabilities	19	22	21	23	26	29	36	41	46	51	58	140
C2. Natural disaster	19	27	27	33	38	42	54	62	70	77	88	203
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio												
Baseline	3	3	3	3	2	2	2	2	3	3	3	8
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2021-2031 2/	3	3	3	3	2	1	1	1	2	2	1	(4)
B. Bound Tests												
B1. Real GDP growth	3	3	3	3	2	2	3	3	4	5	5	14
B2. Primary balance	3	3	3	3	2	2	2	3	3	4	4	9
B3. Exports	3	3	3	3	2	2	2	3	4	4	4	9
B4. Other flows 3/	3	3	3	3	2	2	2	2	3	3	3	8
B5. Depreciation	3	3	3	3	2	2	2	2	3	3	3	8
B6. Combination of B1-B5	3	3	3	3	2	2	2	2	3	4	4	8
C. Tailored Tests												
C1. Combined contingent liabilities	3	3	3	3	2	2	2	2	3	3	4	8
C2. Natural disaster	3	3	3	3	2	2	2	3	4	4	4	9
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the benchmark.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.

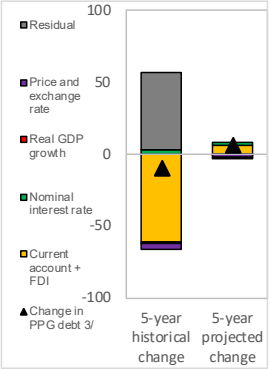
Figure 3. Federated States of Micronesia: Drivers of Debt Dynamics—Baseline Scenario

External Debt

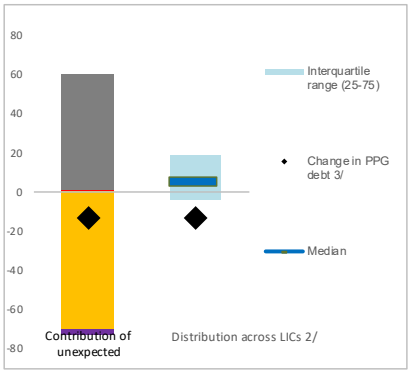
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

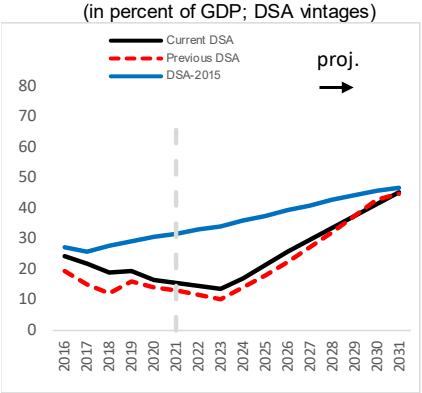


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

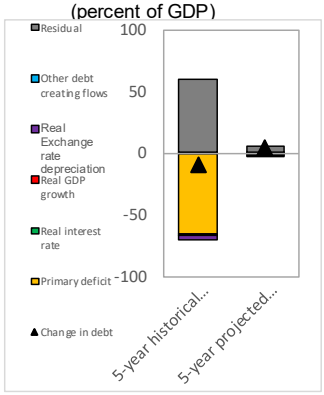


Public Debt

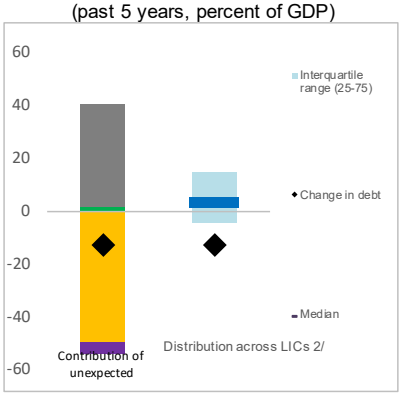
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



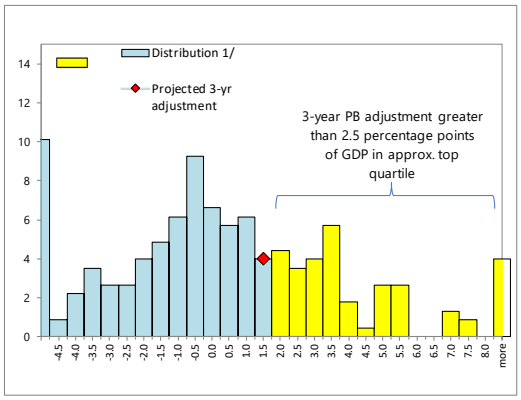
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

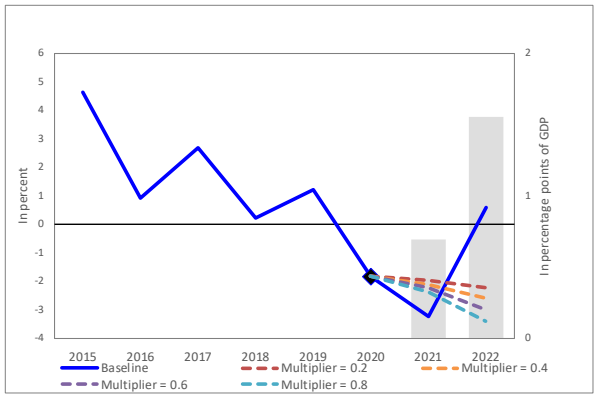
Figure 4. Federated States of Micronesia: Realism Tools 1/

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



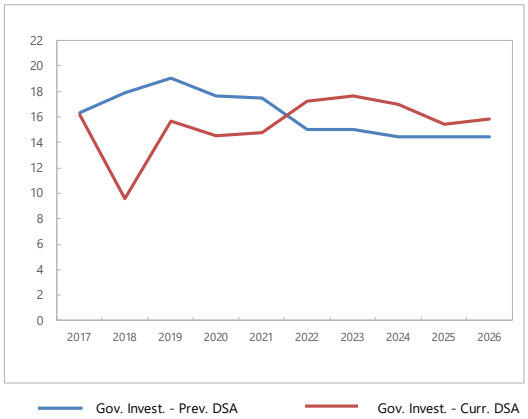
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(percent of GDP)



1/ The data needed to conduct the investment growth tool is not available for the DSA exercise.